

25 July 2026

Editorial Office

Knowledge-Based Systems

Re: Submission of “Evidence contracts for certified knowledge acquisition in retrieval-augmented generation systems”

Dear Editor,

On behalf of all authors, we submit the above manuscript for consideration in *Knowledge-Based Systems*.

The manuscript addresses a knowledge-representation problem that arises when retrieval-augmented generation systems move from evaluation to deployment: how should evidence obligations be represented, and how can a system reason soundly about whether an adaptive acquisition policy satisfies them? We introduce *evidence contracts*, a machine-checkable representation that links answer quality, citation completeness, evidence freshness, and latency to admissible violation rates and a certification confidence. ContractRAG turns this representation into an auditable deployment decision.

The method represents heterogeneous acquisition plans as a rewrite lattice. Risk vouchers encode empirical knowledge about the effects of plan rewrites and compose without independence assumptions. Learned sufficiency models use runtime evidence to choose when a query should acquire stronger knowledge, while a distribution-free fixed-sequence test is the sole source of certification. We prove finite-sample contract validity regardless of candidate count or score quality, characterize cost consistency, and extend the certificate through knowledge drift with an anytime-valid e-process and repeated recertification.

The empirical study covers four heterogeneous knowledge-intensive benchmarks, four large language model families, and two serving backends. Across 1,000 calibration draws, certified policies exceeded their contracts in at most 0.8% of draws under a 10% error budget, whereas uncertified tuning methods reached 64% in the same evaluation. The selected policies reduced model cost by up to 20× relative to the strongest fixed pipeline, or 10–18× after audit costs were included. The released execution matrices and code reproduce the reported results without further model calls.

This contribution fits the journal’s emphasis on knowledge representation, reasoning, and systems that support accountable decisions. Its principal claim is not merely that a RAG pipeline becomes cheaper. It is that explicit knowledge obligations can be represented and used to derive a statistically sound deployment decision; the cost savings follow from that certified reasoning process.

This manuscript is original, has not been published previously, and is not under consideration by another journal. All authors have read and approved the manuscript and agree with its submission to *Knowledge-Based Systems*. The authors declare no competing interests.

Thank you for considering our manuscript. We would be grateful for the opportunity to have it evaluated for publication in *Knowledge-Based Systems*.

Sincerely,

Weiwei Fu and Jinjiang Cui

Co-corresponding authors, on behalf of all authors

Suzhou Institute of Biomedical Engineering and Technology

Chinese Academy of Sciences, Suzhou, China

Email: fuww@sibet.ac.cn; cuijj@sibet.ac.cn